

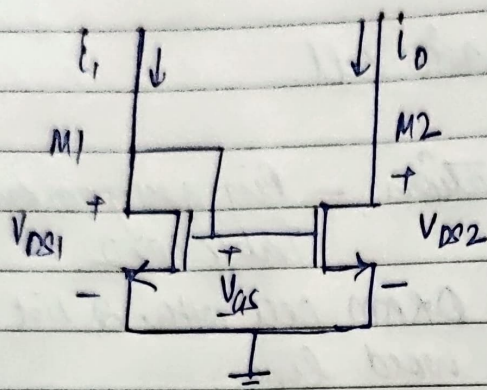
Book Reference: CMOS Analog circuit design.
Allen & Holberg.

Notes: Jasmina Chhikara

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Unit - III

Current Mirror



Principle :- if gate-source potentials of 2 identical MOS are equal, channel currents should be equal.

Input i_1 is mirrored to i_0

M1 is in saturation, $V_{ds1} = V_{gs1}$

Assuming $V_{ds2} \geq V_{gs2} - V_{T2} > V_{T2} \rightarrow M2$ sat

$$\frac{i_0}{i_1} = \left(\frac{L_1 W_2}{W_1 L_2} \right) \left(\frac{V_{gs} - V_{T2}}{V_{gs} - V_{T1}} \right)^2 \left[\frac{1 + \lambda V_{ds2}}{1 + \lambda V_{ds1}} \frac{\left(\frac{K_2'}{K_1'} \right)}{\left(\frac{K_1'}{K_1'} \right)} \right]$$

Since M1 & M2 are identical.

$$\therefore \frac{i_0}{i_1} = \left(\frac{L_1 W_2}{W_1 L_2} \right) \left(\frac{1 + \lambda V_{ds2}}{1 + \lambda V_{ds1}} \right)$$

if $V_{ds2} = V_{ds1}$, then

$$\frac{i_0}{i_1} = \left(\frac{L_1 W_2}{L_2 W_1} \right) \quad (\text{ideal condition})$$

$\therefore$ ratio of aspect
↑

3 effects that cause current mirror to be different from ideal condition :-

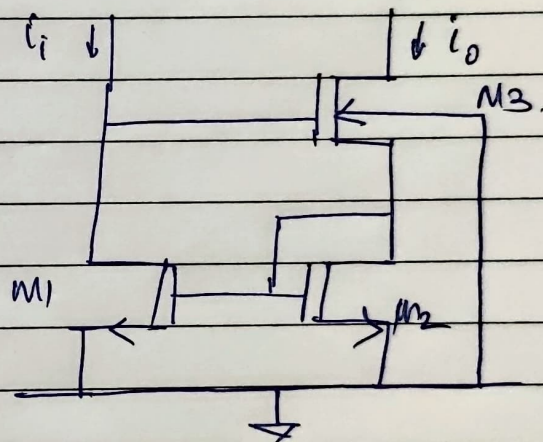
- (i) channel length modulation
- (ii) threshold offset between 2 transistors.
- (iii) imperfect geometrical matching.

Output resistance of a channel current mirror is

$$r_{out} = \frac{1}{g_{ds}} \approx \frac{1}{\lambda I_D}$$

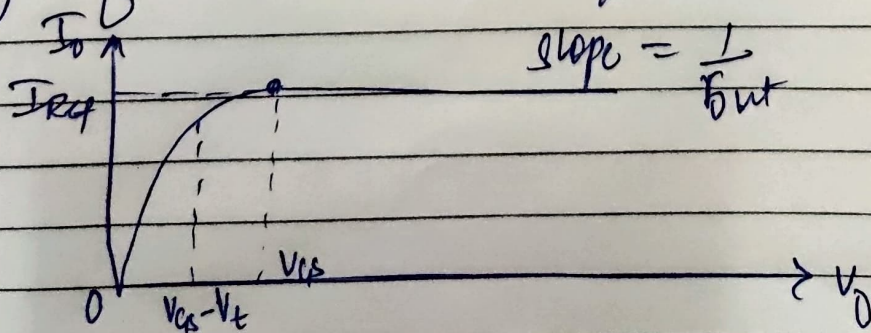
Higher performance current mirrors attempt to increase value of r_{out} .

Wilson current mirror :



The r_{out} is increased using negative current feedback. If i_o increase, $M2$ current increases, mirroring, causes $M1$ to increase.

If i_i constant, resistance from gate of $M3$ to ground, then gate of $M3$ is decreased if i_o increases.



op characteristics of current source.

Op-amp (Operational amplifier)

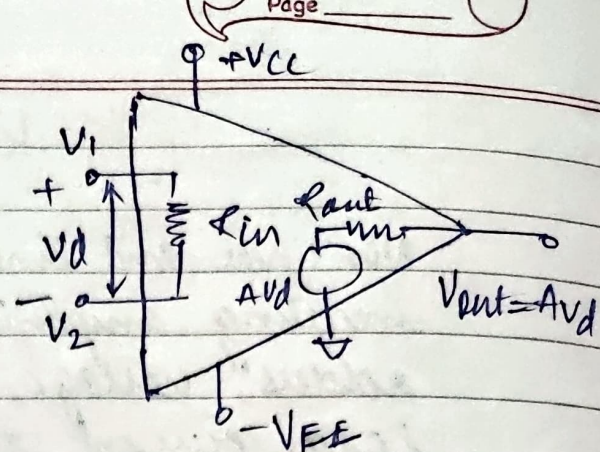
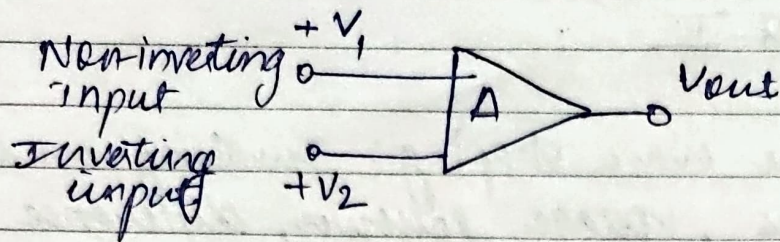
- multistage.
 - high gain
 - direct coupled.
 - negative feedback amplifier.
 - stabilized gain
 - amplify signal from zero Hz to 1 MHz.
- high input impedance ($> 100 \text{ k}\Omega$), low output imp. ($< 100 \Omega$)

Operations performed.

- summation
- subtraction
- multiplication
- differentiation.
- integration
- sign changing
- scale changing
- phase shifting
- voltage regulation
- A to D, D to A converter, V to I, I to V converter.
- sample and hold circuits.

Advantages.

- low cost
- small size
- high reliability
- temperature stability.
- low offset voltage & current.



$$V_{out} = A(V_1 - V_2)$$

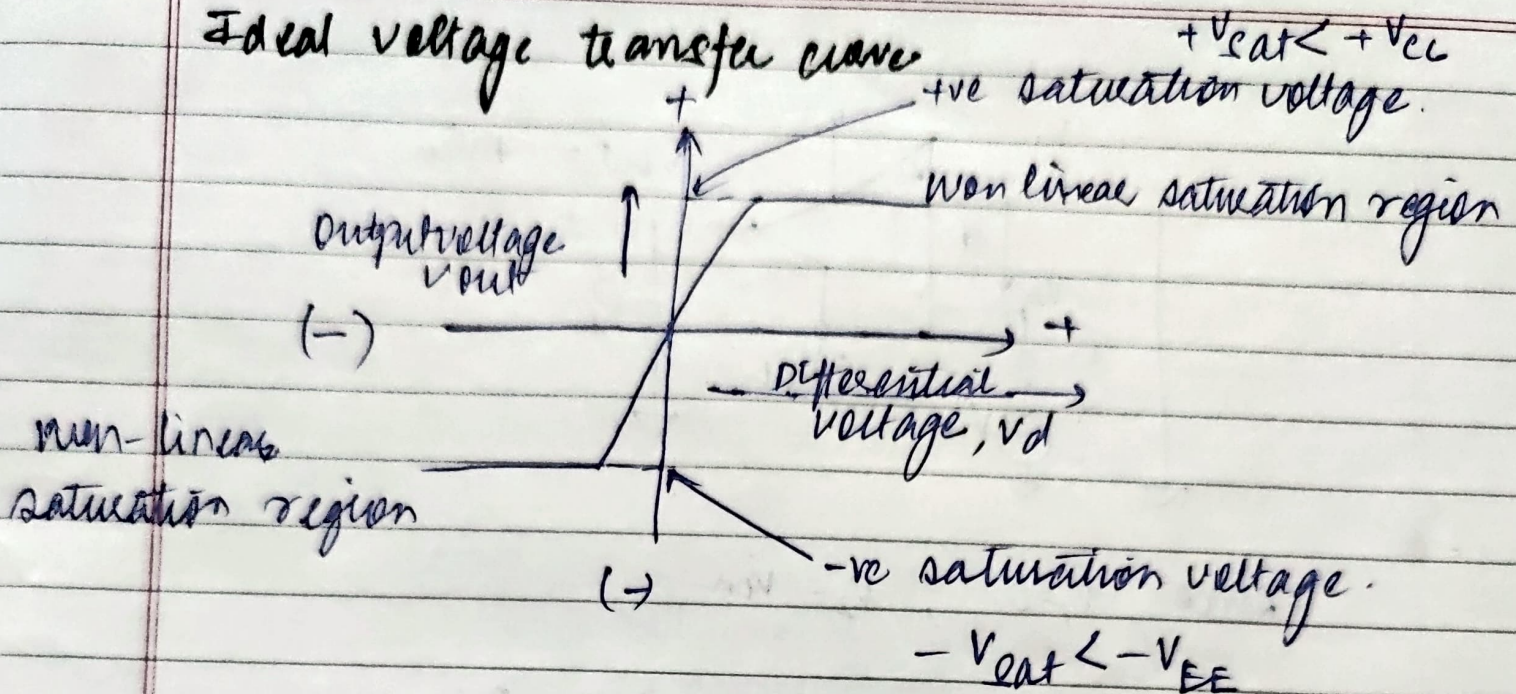
offset null	☐ 1	8	☐ NC (Not connected)
inverting input	☐ 2	7	☐ V+
Non-inverting input	☐ 3	6	☐ output
V-	☐ 4	5	☐ offset null

2 power supply, 2 input terminals, 1 output

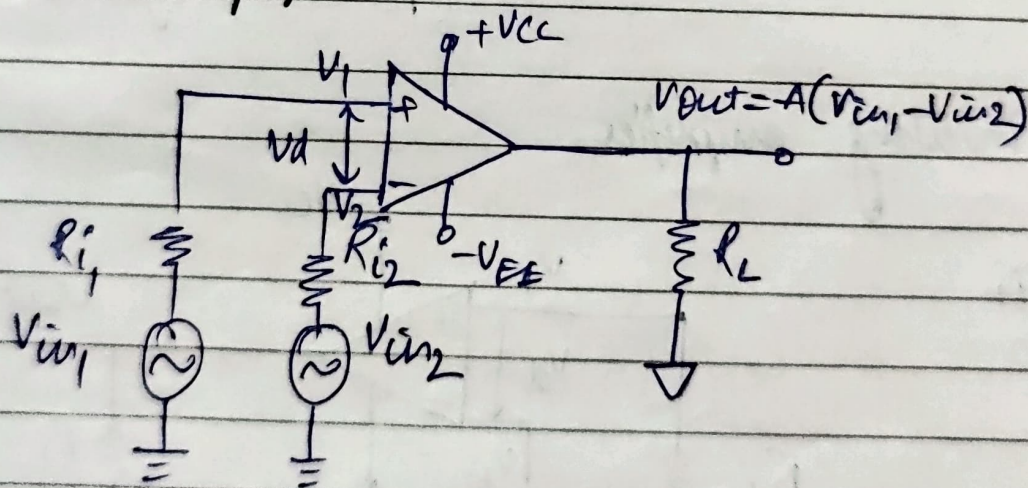
Ideal op-amp

1. open loop gain A is infinite. :-
2. Input resistance R_{in} is infinite.
3. output impedance R_{out} is zero.
4. Perfect balance.
5. Infinite frequency bandwidth.
6. Drift of characteristics with temp. is nil.
7. Common-mode rejection ratio (CMRR) is infinite, so that amplifier is free from undesired commonmode signals.
8. slew rate is infinite : op voltage change occurs simultaneously with input voltage change.
9. output voltage is zero. when input voltage is zero: offset voltage is zero.

Ideal voltage transfer curve



Differential amplifier.



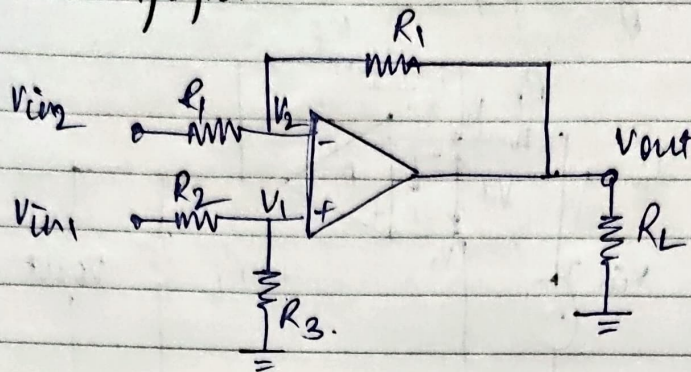
The difference b/w 2 input signals is amplified $\Rightarrow$ differential amplifier.

v_{in1} and v_{in2} can either be ac or dc.

R_{i1} and R_{i2} are negligible as compared to R_{in} .

$$\text{Output voltage} = A(v_{in1} - v_{in2})$$

Differential Amplifier.



No input is grounded $\rightarrow$ reduces amount of noise injected
[since noise in both ends cancel each other in common mode rejection]

Applying superposition theorem in V_{in1} and V_{in2} :-

Inverting : $V_{out1} = -\frac{R_f}{R_1} V_{in2}$ [when $V_{in1} = 0$]

Non-inverting : $V_1 = \frac{R_3}{R_2 + R_3} V_{in1}$ [when $V_{in2} = 0$]
then

$$V_{out1} = \left(1 + \frac{R_f}{R_1}\right) V_1 = \left(1 + \frac{R_f}{R_1}\right) \left(\frac{R_3}{R_2 + R_3}\right) V_{in1}$$

if $R_1 = R_2$ and $R_f = R_3$

$$V_{out1} = \left(\frac{R_f(R_2 + R_3)}{R_1(R_2 + R_3)}\right) V_{in1} = \frac{R_f}{R_1} V_{in1}$$

Net output, $V_{out} = V_{out1} + V_{out2}$

$$= \frac{R_f}{R_1} V_{in1} - \frac{R_f}{R_1} V_{in2}$$

$$= -\frac{R_f}{R_1} (V_{in2} - V_{in1})$$

Differential gain, $A_d = \frac{V_{out}}{V_{in2} - V_{in1}} = \frac{-R_f}{R_1}$

D.A can operate in 3 different modes:-

① Single ended mode :- input is applied to one of the two inputs.

② Differential mode :-

$$V_{in2} = -V_{in1}$$

$$\therefore V_{in} = V_{in2} - V_{in1} = V_{in2} - (-V_{in2}) = 2V_{in2}$$

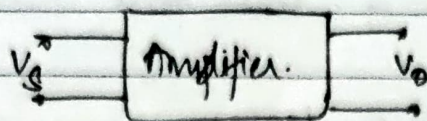
$$V_{out} = \frac{-R_f}{R_1} [V_{in2} - V_{in1}] = \frac{-2R_f}{R_1} V_{in2}$$

③ Common mode :-

$$V_{in} = V_{in2} - V_{in1} = 0$$

Output voltage = 0.

Feedback in Amplifiers. NPTEL.



open loop amp^r : Having no feedback.

$$\text{Gain} = \frac{V_o}{V_i} \quad V_o = A \cdot V_i$$

If A change to $A + \Delta A$ [for any reason]
Output $V_o = (A + \Delta A) V_i$
changes

Not acceptable

Gain may change due to :

- ① Temperature change
- ② Aging
- ③ Replacement of device
- ④ Voltage fluctuations

$\therefore$ feedback incorporated. e- function of op mixed with g/p
- brings stability.

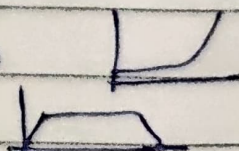
Two types of feedback :

① Input = $V_i + V_f$ [positive feedback, reduces amp^r
(sum will increase always) stability] used only in
oscillator ckts.

② Input = $V_i - V_f$ [Negative feedback]

- * increase stability of gain
- * reduces non-linear distortion
- * Increase the Bandwidth

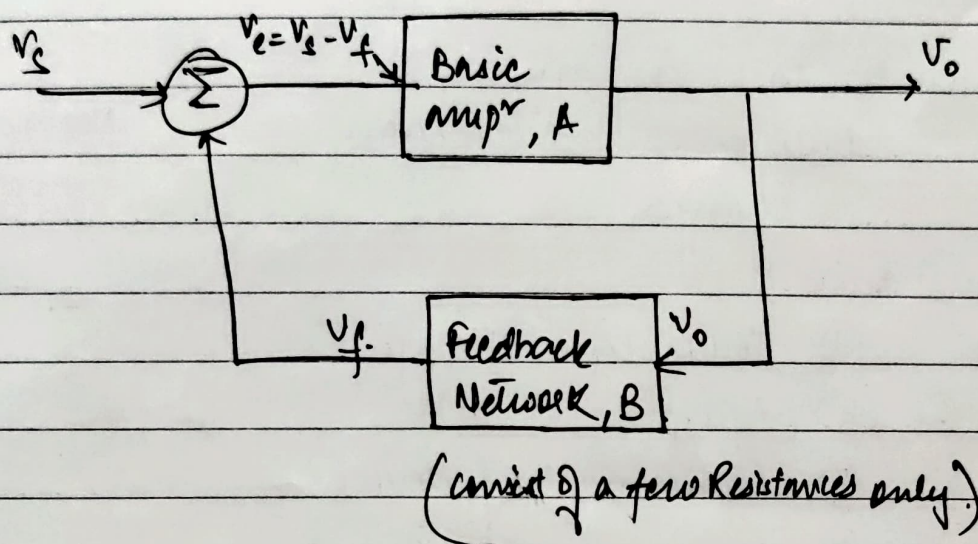
* modifies (increase or decrease) the I/P or O/P impedance



Disadvantage of neg. feedback - Gain falls.

⊛ But we can add additional stage to multiply gain [multistage amp^r]

GENERAL FEEDBACK RELATION.



Openloop gain, $A = \frac{V_o}{V_e}$

Feedback factor, $B = \frac{V_f}{V_o}$

V_e = Error voltage. [it should be minimum]

$$V_o = A \cdot V_e \quad \text{--- (1)}$$

$$V_e = V_s - V_f \quad \text{--- (2)}$$

$$V_f = B \cdot V_o \quad \text{--- (3)}$$

Substitute (3) into (2), (2) into (1).

$$V_o = A (V_s - B \cdot V_o)$$

$$V_o = AV_s - AB \cdot V_o$$

$$V_o + AB \cdot V_o = AV_s$$

$$V_o (1 + AB) =$$

$$V_o = A (V_s - BV_o)$$

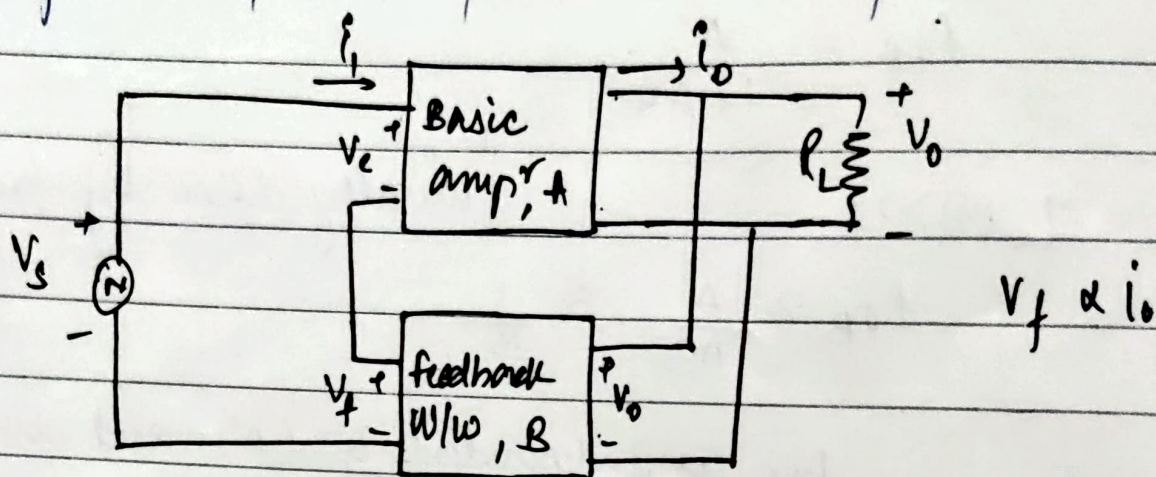
$$V_o = AV_s - ABV_o$$

$$V_o (1 + AB) = AV_s$$

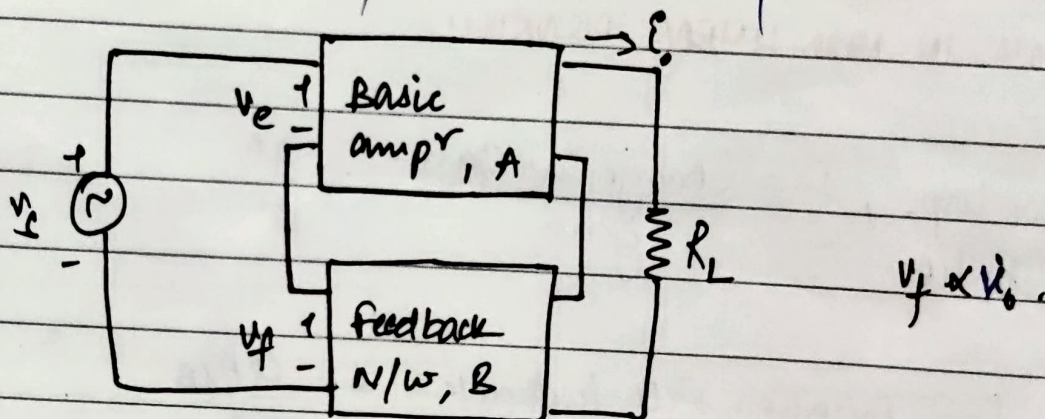
$$\boxed{\frac{V_o}{V_s} = \frac{A}{1 + AB} = A_f}$$

DIFFERENT TYPES OF NEGATIVE FEEDBACK.

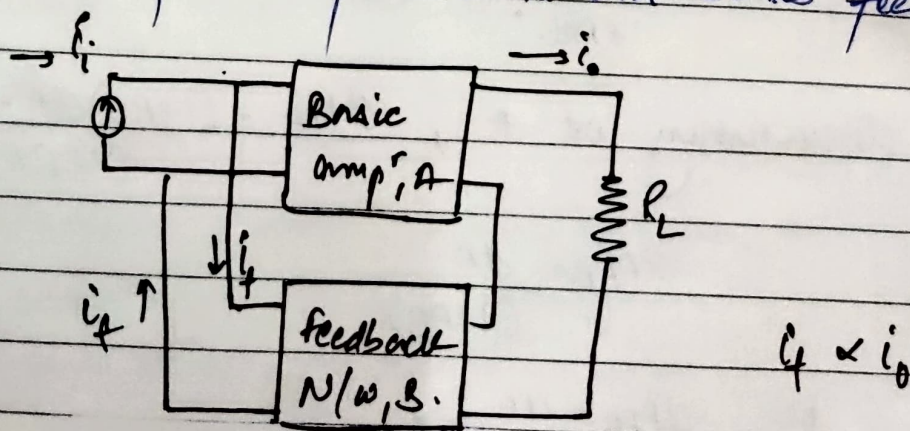
① Voltage series feedback / Series-shunt feedback.



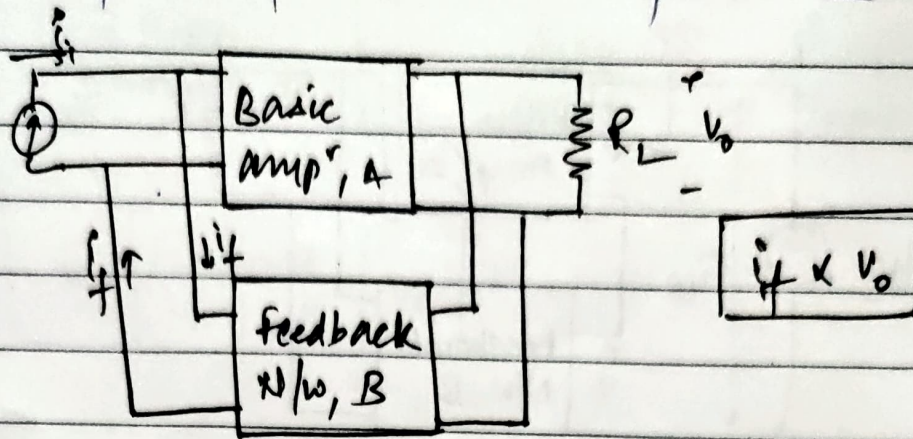
② Current series feedback / Series-series feedback



③ Current shunt feedback / Shunt-shunt-series feedback



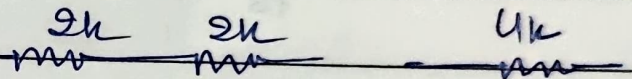
① Voltage Shunt feedback / Shunt-Shunt feedback



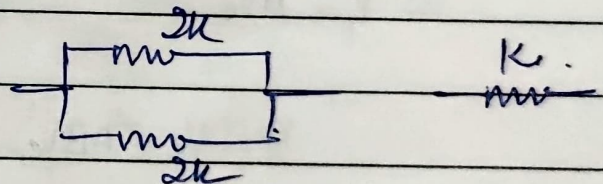
EFFECT OF NEGATIVE FEEDBACK ON IP AND OP IMPEDANCE

$$Z_i(FB) = (1 + BA) Z_i$$

(if input is series.)



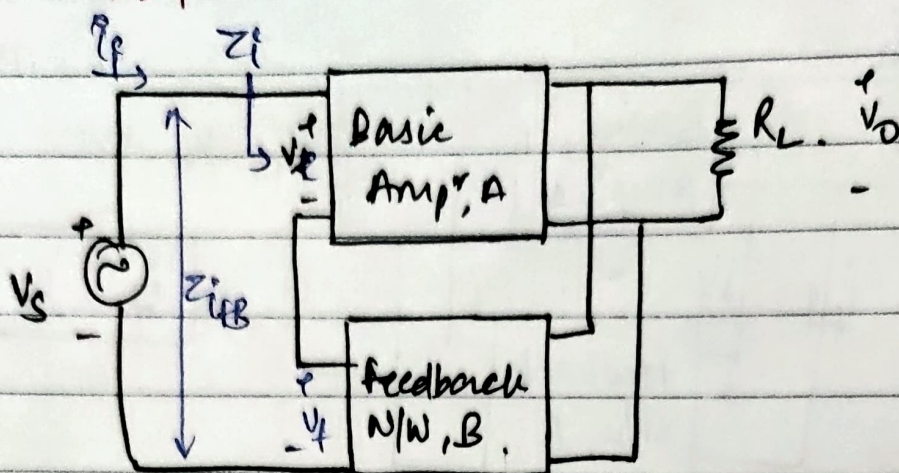
$$Z_i(FB) = \frac{Z_i}{1 + AB}$$



(if input is shunt resistance)

* Similar applicable for output

VOLTAGE SERIES NEGATIVE FEEDBACK



INPUT IMPEDANCE

$$Z_i = \frac{V_e}{i_i} \quad [\text{open loop impedance}]$$

$$Z_{ifb} = \frac{V_s}{i_i}$$

$$V_s = V_e + B V_o$$

$$\text{as } V_f = B V_o$$

$$\text{as } V_o = A V_e$$

$$V_s = V_e + A B V_e$$

$$V_s = V_e (1 + A B)$$

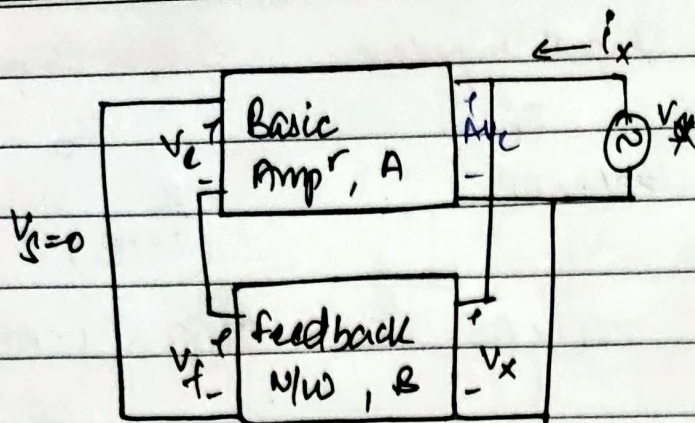
Divide both sides by i_i

$$\frac{V_s}{i_i} = \frac{V_e}{i_i} (1 + A B)$$

$$\boxed{Z_{ifb} = Z_i (1 + A B)}$$

OUTPUT IMPEDANCE

R_L is not included while calculating output impedance.
(Basic principle)



V_i applied (test signal) to op.
Input voltage set to 0.

$$Z_{ofB} = \frac{V_x}{i_x}$$

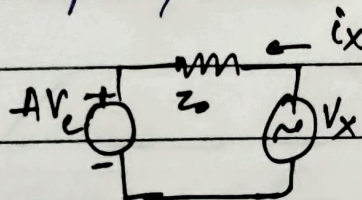
Since, $V_i = 0$, $V_e = -V_f$.

and $V_f = BV_x$

$$\therefore V_e = -BV_x$$

from the equivalent ckt for output port.

$$i_x = \frac{V_x - AV_e}{Z_o}$$



$$i_x = \frac{V_x + ABV_x}{Z_o}$$

$$i_x = \frac{V_x(1+AB)}{Z_o}$$

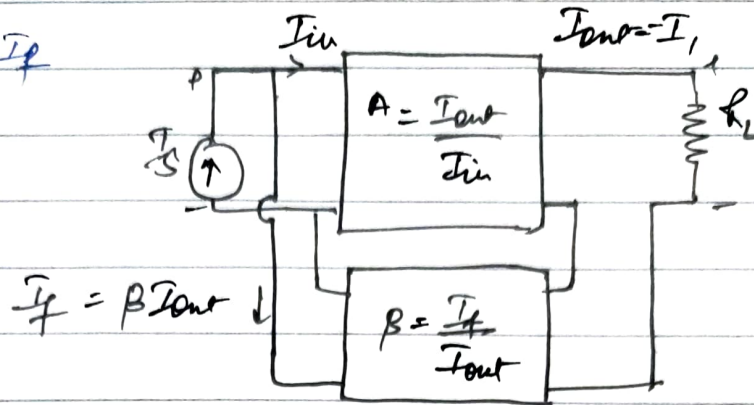
$$Z_o = \frac{V_x(1+AB)}{i_x}$$

$$Z_{ofB} = \frac{Z_o}{(1+AB)}$$

CURRENT SHUNT FEEDBACK

INPUT IMPEDANCE

$$I_s = I_{in} + I_f$$



$$I_f = \beta I_{out}$$

$$I_s = I_{in} + \beta I_{out}$$

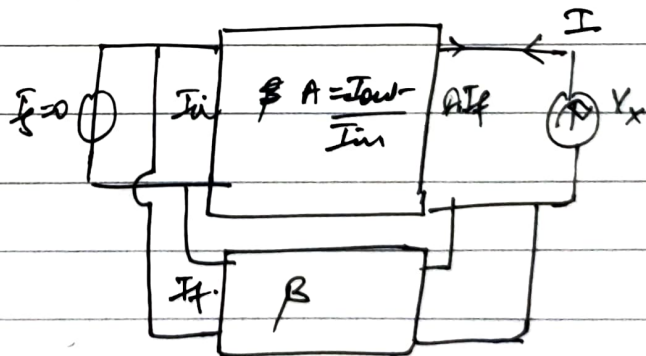
$$= I_{in} + \beta (A I_{in})$$

$$I_s = I_{in} (1 + \beta A)$$

$$Z_{inf} = \frac{V_{in}}{I_s} = \frac{V_{in}}{I_{in} (1 + \beta A)} = \frac{Z_{in}}{1 + \beta A}$$

OUTPUT IMPEDANCE

with $I_s = 0$, $V_{in} = -I_f = -\beta I_{out} = -\beta I$



$$I = \frac{V}{Z_{out}} - A I_f$$

$$I = \frac{V}{Z_{out}} - A \beta I$$

$$\frac{V}{Z_{out}} = I + \beta A I = I (1 + \beta A)$$

$$\frac{V}{I} = Z_{out} (1 + \beta A)$$

$$\Rightarrow Z_{outf} = Z_{out} (1 + \beta A)$$